# Intro to Machine Learning
# Quiz 3 Practice Problems
Practice problems, not turned in

**Directions:** These practice problems cover the material for Quiz 3: linear discriminant analysis in Part 1, quadratic discriminant analysis in Part 2, and basis functions and splines in Part 3. They are not turned in, but the quiz draws on them, so work through each one before looking at the solutions. Problems adapted from ISL cite the exercise; see the book for the original wording.

## Part 1: Linear Discriminant Analysis

Notation follows ISL 4.4. Classes are $k = 1, \dots, K$, $\pi_k = \Pr(Y = k)$ is the prior, $f_k(x)$ is the density of $X$ in class $k$, and $p_k(x) = \Pr(Y = k \mid X = x)$ is the posterior. When fitting LDA by hand, use the pooled variance with divisor $n - K$.

1. **The LDA discriminant.** (ISL 4.8, Exercise 2; ISL 4.4.1–4.4.2, ESL 4.3.)

   (a) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma^2)$, with the same $\sigma^2$ in every class. Show that the Bayes classifier assigns $x$ to the class with the largest

   $$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k.$$

   (b) With $K = 3$ and $p = 4$, how many parameters does LDA estimate (priors, means and the shared covariance)? How many coefficients does multinomial logistic regression estimate? The LDA posterior has the same form as a multinomial logistic regression, so what is LDA doing with the extra parameters?

2. **LDA with one predictor.** (ISL 4.8, Exercise 7; ISL 4.4.2.) We want to predict whether a stock will issue a dividend this year (yes or no) from $X$, last year's percentage profit. Among companies that issued a dividend, $X$ averaged $\bar{x} = 10$; among those that didn't, $\bar{x} = 0$. The variance of $X$ in both groups is $\hat{\sigma}^2 = 36$, and 80% of companies issued dividends. Assume $X$ is normal in each group. Find the probability that a company with $X = 4$ issues a dividend this year.

3. **Fitting LDA by hand.** (ISL 4.4.2.) A training set has one predictor:

   $$\text{class 1: } x = 1, 2, 3; \qquad \text{class 2: } x = 5, 6, 6, 7.$$

   (a) Compute $\hat{\pi}_k$, $\hat{\mu}_k$ and the pooled variance $\hat{\sigma}^2$.

   (b) Find the LDA decision boundary. Which way did the unequal priors move it from the midpoint of the means?

## Part 2: Quadratic Discriminant Analysis

QDA assumes $X \mid Y = k \sim N(\mu_k, \Sigma_k)$; naive Bayes assumes $f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$.

4. **QDA is quadratic.** (ISL 4.8, Exercise 3; ISL 4.4.3, ESL 4.3.) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$, with a different variance in each class. Find the discriminant $\delta_k(x)$ the Bayes classifier maximizes, and show that the decision boundary between two classes is not linear in $x$ unless $\sigma_1 = \sigma_2$.

5. **QDA by hand: same means, different spreads.** (ISL 4.4.3.) Two equally likely classes have $X \mid Y = 1 \sim N(0, 1)$ and $X \mid Y = 2 \sim N(0, 4)$.

   (a) Find the Bayes decision rule. Where is each class predicted?

   (b) Now let $\pi_1 = 0.8$. How does the rule change?

6. **LDA or QDA?** (ISL 4.8, Exercise 5; ISL 4.4.3, 4.5.) Answer each and explain.

   (a) If the Bayes decision boundary is linear, do we expect LDA or QDA to do better on the training set? On the test set?

   (b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to do better on the training set? On the test set?

   (c) In general, as $n$ increases, do we expect the test accuracy of QDA relative to LDA to improve, decline or stay the same?

   (d) True or false: even if the Bayes boundary is linear, QDA will probably achieve a lower test error rate than LDA, because QDA is flexible enough to model a linear boundary.

   (e) With $p = 10$ predictors and $K = 3$ classes, how many covariance parameters do LDA, QDA and Gaussian naive Bayes each estimate? How many observations does each class need for QDA's $\hat{\Sigma}_k$ to be invertible?

## Part 3: Basis Functions and Splines

Here $I(\cdot)$ is the indicator, equal to 1 when its condition holds and 0 otherwise, and $g^{(m)}$ is the $m$th derivative of $g$, with $g^{(0)} = g$.

7. **Penalizing different derivatives.** (ISL 7.9, Exercise 2; ISL 7.5.) A curve $\hat{g}$ is fitted to $n$ points by

$$\hat{g} = \arg\min_{g} \Big( \sum_{i=1}^{n} \big(y_i - g(x_i)\big)^2 + \lambda \int \big[g^{(m)}(x)\big]^2 \, dx \Big).$$

   Sketch an example of $\hat{g}$ for each of the following.

   (a) $\lambda = \infty$, $m = 0$.

   (b) $\lambda = \infty$, $m = 1$.

   (c) $\lambda = \infty$, $m = 2$.

   (d) $\lambda = \infty$, $m = 3$.

   (e) $\lambda = 0$, $m = 3$.

8. **Sketching a fit from its basis functions, I.** (ISL 7.9, Exercise 3; ISL 7.3.) A curve is fitted with basis functions $b_1(X) = X$ and $b_2(X) = (X-1)^2 I(X \geq 1)$. The regression $Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon$ gives $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$ and $\hat{\beta}_2 = -2$. Sketch the fitted curve for $-2 \leq X \leq 2$, marking the intercepts, slopes and anything else of interest.

9. **Sketching a fit from its basis functions, II.** (ISL 7.9, Exercise 4; ISL 7.2, 7.3.) A curve is fitted with basis functions

$$b_1(X) = I(0 \leq X \leq 2) - (X-1) I(1 \leq X \leq 2),$$

$$b_2(X) = (X-3) I(3 \leq X \leq 4) + I(4 < X \leq 5).$$

The regression $Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon$ gives $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$ and $\hat{\beta}_2 = 3$. Sketch the fitted curve for $-2 \leq X \leq 6$, marking the intercepts, slopes and anything else of interest.

10. **Penalizing the third or the fourth derivative?** (ISL 7.9, Exercise 5; ISL 7.5.) Consider

$$\hat{g}_1 = \arg\min_g \Big( \sum_{i=1}^{n} \big(y_i - g(x_i)\big)^2 + \lambda \int \big[g^{(3)}(x)\big]^2 \, dx \Big),$$

$$\hat{g}_2 = \arg\min_g \Big( \sum_{i=1}^{n} \big(y_i - g(x_i)\big)^2 + \lambda \int \big[g^{(4)}(x)\big]^2 \, dx \Big).$$

   (a) As $\lambda \to \infty$, which of $\hat{g}_1$ and $\hat{g}_2$ has the smaller training RSS?

   (b) With $\lambda = 0$, which has the smaller training RSS and test RSS?